

## **M1 - Financial Economics, academic year 2025 - 2026**

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## **FINANCIAL ECONOMETRICS WITH PYTHON**

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## Question 1

### Preliminary Step

| Name  | Ticker | Length                              | Initial Date | End Date   | Stock Market      | Country |
|-------|--------|-------------------------------------|--------------|------------|-------------------|---------|
| CAC40 | FCHI   | 30 years<br>360 Months<br>7639 Days | 31/12/1994   | 31/12/2024 | Euronext<br>Paris | France  |

|                                   | Daily      | Monthly  | Annual   |
|-----------------------------------|------------|----------|----------|
| <b>Mean (%)</b>                   | 0.0177     | 0.3908   | 4.6990   |
| <b>Std. Deviation (%)</b>         | 1.3682     | 5.2350   | 20.7603  |
| <b>Diameter C.I. Mean (%)</b>     | 0.0307     | 0.5415   | 7.5560   |
| <b>Skewness</b>                   | -0.2026    | -0.5102  | -1.0115  |
| <b>Kurtosis</b>                   | 9.0593     | 4.1513   | 4.0454   |
| <b>Excess Kurtosis</b>            | 6.0593     | 1.1514   | 1.0454   |
| <b>Minimum (%)</b>                | -13.0983   | -19.2254 | -55.6527 |
| <b>5% Quantile (%)</b>            | -2.1549    | -8.8186  | -34.6230 |
| <b>25% Quantile (%)</b>           | -0.6407    | -2.7118  | -3.0914  |
| <b>Median (%)</b>                 | 0.0558     | 0.9961   | 8.8560   |
| <b>75% Quantile (%)</b>           | 0.7190     | 3.6520   | 20.1498  |
| <b>95% Quantile (%)</b>           | 2.0454     | 8.3264   | 26.7580  |
| <b>Maximum (%)</b>                | 10.5946    | 18.3312  | 41.2933  |
| <b>Jarque–Bera Stat</b>           | 11736.8594 | 35.4010  | 6.2655   |
| <b>Jarque–Bera p-value (×100)</b> | 0.0000     | 0.0000   | 4.3597   |
| <b>Lilliefors Stat</b>            | 0.0689     | 0.0581   | 0.1148   |
| <b>Lilliefors p-value (×100)</b>  | 0.1000     | 0.7944   | 42.4372  |
| <b>N. obs</b>                     | 7638.0000  | 359.0000 | 29.0000  |

Figure 1 : Descriptive Statistics of CAC 40 Log>Returns (Daily, Monthly, and Annual)

## Stylized facts of financial returns

### Stylized Fact 2: Returns are stationary

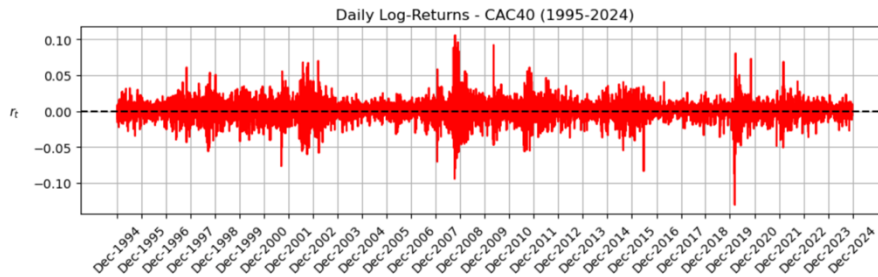


Figure 2: Log returns: daily “adjusted closing” of CAC40 from Yahoo.com. Sample: 31-01-1994 to 31-12-2024. Dashed horizontal line is at level 0.

It can be seen in figure 2 that the most recent critical difference happened in 2020 during the first year of the pandemic. But over the last 20 years, the returns fluctuate around a mean close to zero, since there is no apparent upward or downward trend. Which goes with the idea of the stylised fact 2 about the returns being stationary.

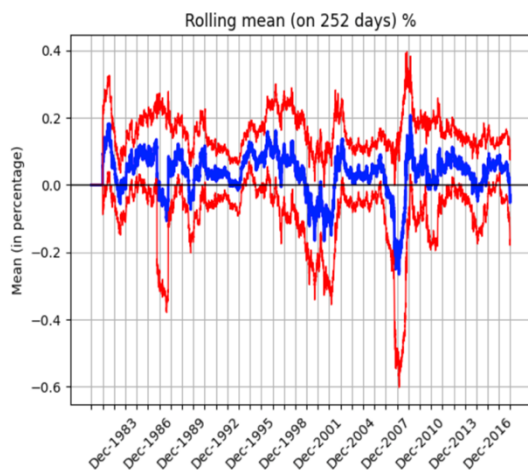


Figure 3: Rolling 1 year mean for  $r_t$ , the daily percentage log-return (computed over the previous 252 days) from the “adjusted closing” of CAC40. from Yahoo.com. Sample: 29-12-1981 to 31-12-2018.

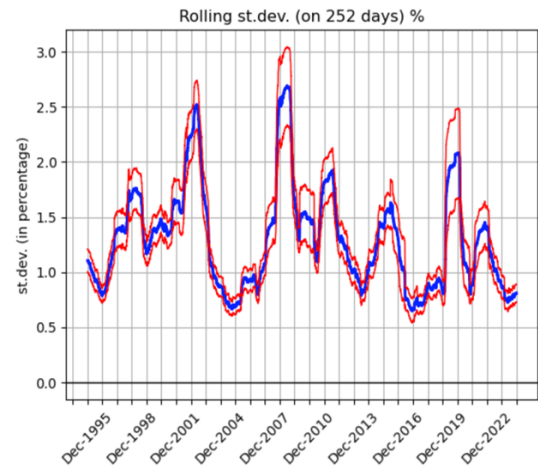


Figure 4: Rolling 1 year percentage standard deviation for  $r_t$ , the daily log-return (computed over the previous 252 days) from the “adjusted closing” of CAC40. from Yahoo.com. Sample: 29-12-1981 to 31-12-2018.

It can be observed in figure 3 that the rolling mean stays close to zero besides some extremes. Which reflects stationarity. And that the rolling standard deviation in figure 4 shows the volatility of the CAC40. High volatility can be explained by crises and low volatility reflects a stable market. But looking at the all 252 days, the range of variation stays approximately similar. Which also shows a stationarity in the returns of the CAC40.

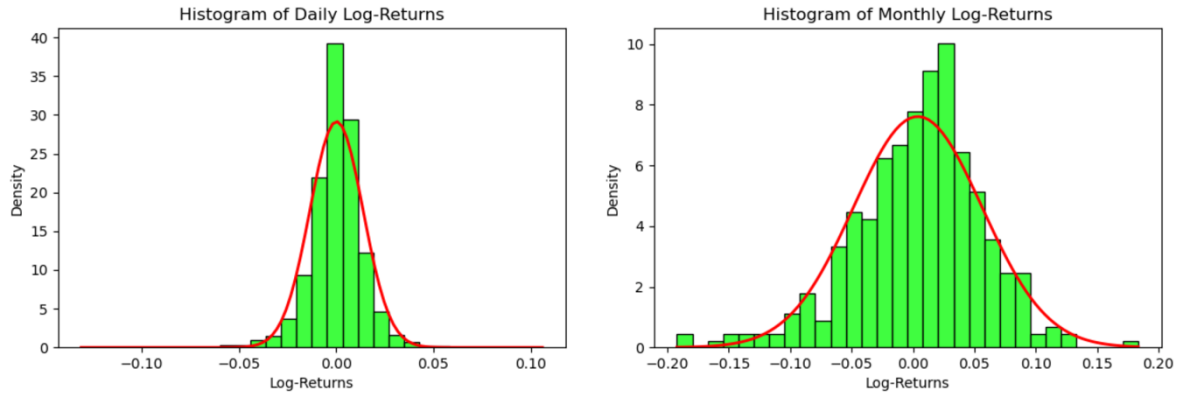


Figure 5: Log returns: histograms of daily, monthly “adjusted closing” of CAC40 from Yahoo.com.Sample: 31-12-1994 to 31-12-2024.

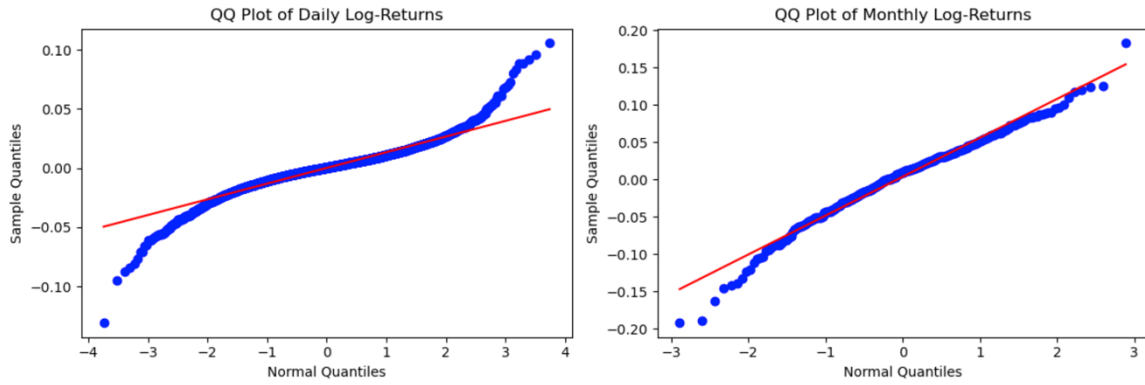


Figure 6: QQ plot against quantiles of normal distribution with same mean and variance as the empirical distribution of returns. QQ plots of daily, monthly “adjusted closing” of CAC40 from Yahoo.com.Sample: 31-12-1994 to 31-12-2024

Finally, it can be seen on the histograms (figure 5) for daily and monthly log-returns that the highest density is around zero, meaning that most returns are concentrated around zero, indicating that the mean is approximately constant and close to zero. Moreover, the red line shows the normal distribution of the time series, and since some extreme returns exceed the red line, the time series is non-normal. Similarly, the QQ Plots (figure 6) of the daily and monthly log-returns show the QQ plots comparing the empirical quantiles of returns to those of a theoretical normal distribution. The points near the center lie on the red line, meaning that the returns near the mean almost behave like a normal distribution and that the distribution is centered and stable. But fat-tails can still be observed, meaning that some blue dots do not lie on the red line.

### Stylized Fact 3: Returns are asymmetric

The distribution of returns is asymmetric and often negatively skewed, reflecting the fact that the downturns of financial markets are often much steeper than the recoveries. Investors tend to react more strongly to negative news than to positive news.

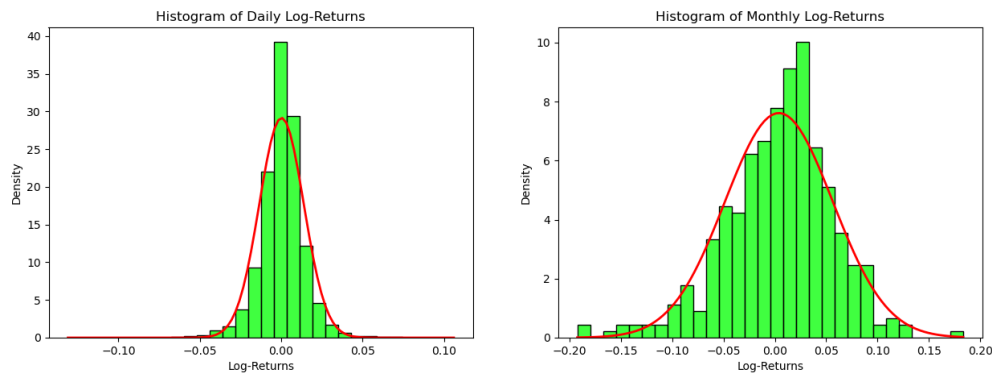


Figure 3

On the left, the histogram of daily log-returns for the CAC40 is highly concentrated around zero but shows fat tails and slight asymmetry. This indicates that extreme positive or negative returns occur more frequently than what would be expected under a normal distribution. On the right graph, the monthly log-returns histogram looks wider and somewhat closer to normal yet still displays asymmetry and excess kurtosis. CAC40 returns are not perfectly normal. They exhibit fat tails and a slight skewness, consistent with the third stylized fact of financial markets.

### Stylized Fact 4: Returns have heavy tails

The distribution of returns often exhibits heavier tails than those of a Gaussian distribution, that is returns are leptokurtic.

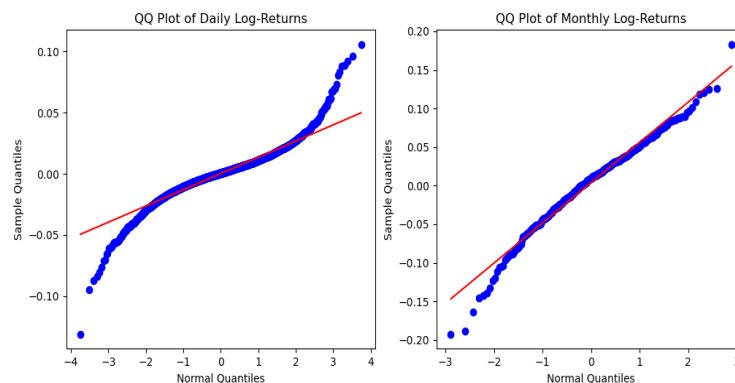


Figure 4

The QQ Plot of Daily Log-Returns provides clear graphical evidence that the tails of the empirical distribution are heavier than those of the normal distribution. Left Tail (Lower

Quantiles): The points on the left side of the graph (representing the large negative returns) lie below the straight red line. This implies that the large negative returns observed in the sample are much smaller (more negative) than what you would expect from a Normal random variable with the same empirical mean and standard deviation. This confirms that the left tail of the empirical distribution is heavier. Right Tail (Upper Quantiles): Similarly, the points on the right side of the graph (representing the large positive returns) lie above the straight red line. This implies that the large positive returns observed in the sample are much larger than expected under the Normal distribution. This confirms the right tail is also heavier. This pronounced "S" shape deviation from the straight line in the QQ plot's tails is the graphical signature of leptokurtosis and provides definitive evidence for Stylized Fact 4.

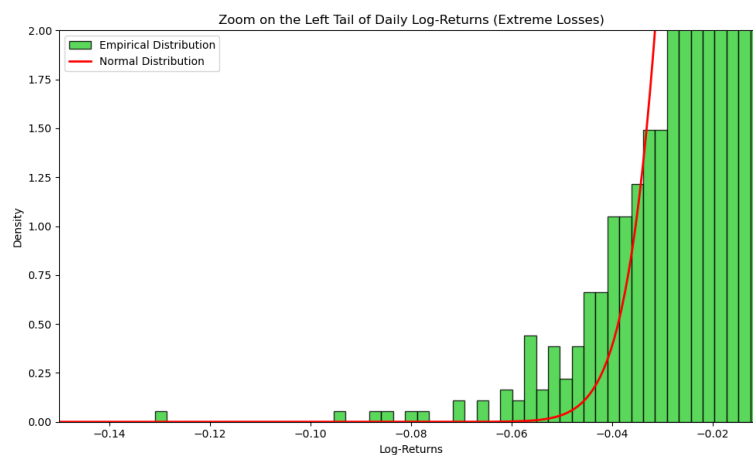


Figure 5

Zooming in on the extreme values of the Histogram (the tails) shows that the real market returns still have some height where the simple Normal curve has already dropped almost to zero. This clearly means that days with very big gains or very big losses happen more often than the Normal distribution expects which is also an evidence for Stylised Fact 4.

### Stylized Fact 5: Aggregational Gaussianity

The Stylized Fact of Aggregational Gaussianity states that the distribution of returns tends to approximate a Normal (Gaussian) distribution as the frequency of the data is aggregated over longer time horizons. This phenomenon is consistent with the Central Limit Theorem (CLT).

The clear and progressive decrease in Excess Kurtosis with aggregation is an evidence for Aggregational Gaussianity. It falls from a high value of 6.0593 at the daily level to a much lower value of 1.1514 at the monthly level, and 1.0454 at the annual level. This convergence toward zero (the Normal value) is the statistical signature of Aggregational Gaussianity.

|          |        |        |        |
|----------|--------|--------|--------|
| Kurtosis | 6.0593 | 1.1514 | 1.0454 |
|----------|--------|--------|--------|

The comparison of the QQ Plots (Fig.6) visually confirms this trend. The pronounced "S" shape deviation visible in the daily quantiles (evidence of heavy tails) is significantly attenuated in the monthly quantiles. The monthly points are visibly more aligned with the straight reference line, indicating that the distribution has lost a significant portion of its leptokurtosis.

In conclusion, although the daily returns are highly non-normal, aggregating the data over longer horizons tends to "normalize" the distribution, which is in perfect agreement with Stylized Fact 5.

### Stylized Fact 6: Returns are not autocorrelated

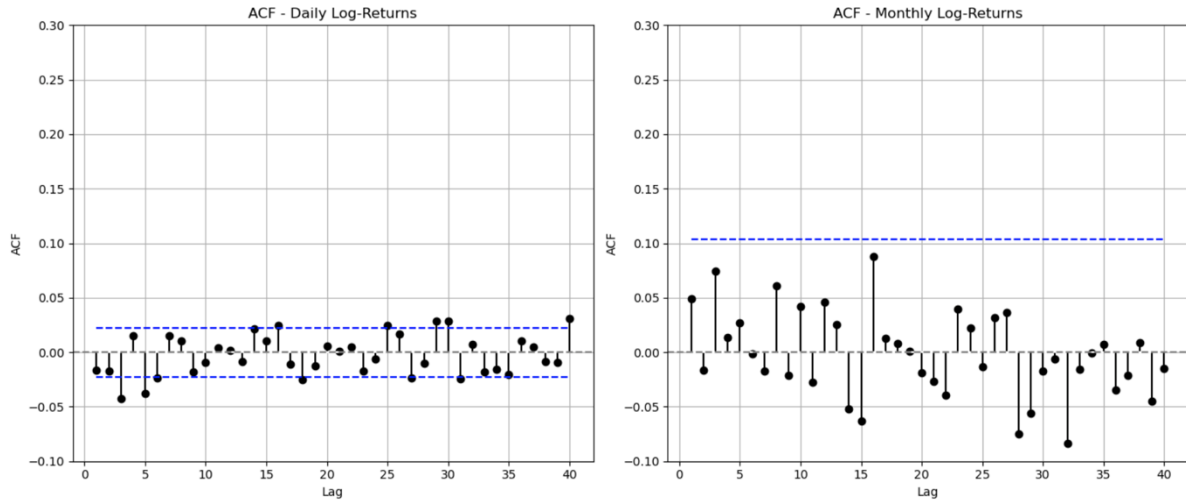


Figure 6: Empirical Autocorrelation (ACF) of the daily and monthly log-return from the "adjusted closing" of CAC40. The autocorrelation of order 0 is not reported to have a better graphical representation of the smaller autocorrelations of order  $k$ . Sample: 31-12-1994 to 31-12-2024. Data source: Yahoo.com

On these diagrams the correlation between 40 different lags can be seen.

For example, on the monthly ACF, if we take lag  $k=15$ , it means that we took the correlation between a log return  $r_t$  and  $r_{t-15}$ .

The autocorrelation, denoted  $\rho(k)$ , of a weak stationary process  $r_t$  is the correlation between values of the process at different times, defined as:  $\rho_k = \text{Corr}(r_t, r_{t-k})$ .

Looking at the daily log-returns ACF plot, The height of the bar represents the correlation coefficient at that lag. Thus, since at multiple lags the bar extends beyond the significance bounds, it indicates significant autocorrelation at that lag. While for the monthly log-returns there is not significant autocorrelation.

|         | ACF ( $\rho_k$ ) | ACF diam. | ACF t-test | Box–Pierce stat | BP pval | Ljung–Box stat | LB pval | $\chi^2_{\{k,0.95\}}$ |
|---------|------------------|-----------|------------|-----------------|---------|----------------|---------|-----------------------|
| lag (k) |                  |           |            |                 |         |                |         |                       |
| 1       | -0.017           | 0.022     | -1.469     | 2.157           | 0.142   | 2.158          | 0.142   | 3.841                 |
| 2       | -0.017           | 0.022     | -1.529     | 4.496           | 0.106   | 4.498          | 0.106   | 5.991                 |
| 3       | -0.043           | 0.022     | -3.728     | 18.393          | 0.000   | 18.405         | 0.000   | 7.815                 |
| 4       | 0.015            | 0.022     | 1.338      | 20.184          | 0.000   | 20.196         | 0.000   | 9.488                 |
| 5       | -0.038           | 0.022     | -3.297     | 31.057          | 0.000   | 31.079         | 0.000   | 11.070                |
| 6       | -0.023           | 0.022     | -2.036     | 35.201          | 0.000   | 35.228         | 0.000   | 12.592                |
| 7       | 0.015            | 0.022     | 1.303      | 36.899          | 0.000   | 36.928         | 0.000   | 14.067                |
| 8       | 0.010            | 0.022     | 0.899      | 37.707          | 0.000   | 37.737         | 0.000   | 15.507                |
| 9       | -0.018           | 0.022     | -1.558     | 40.134          | 0.000   | 40.168         | 0.000   | 16.919                |
| 10      | -0.009           | 0.022     | -0.822     | 40.810          | 0.000   | 40.845         | 0.000   | 18.307                |
| 11      | 0.004            | 0.022     | 0.326      | 40.917          | 0.000   | 40.951         | 0.000   | 19.675                |
| 12      | 0.002            | 0.022     | 0.150      | 40.939          | 0.000   | 40.974         | 0.000   | 21.026                |
| 13      | -0.009           | 0.022     | -0.748     | 41.498          | 0.000   | 41.534         | 0.000   | 22.362                |
| 14      | 0.021            | 0.022     | 1.860      | 44.957          | 0.000   | 45.001         | 0.000   | 23.685                |
| 15      | 0.010            | 0.022     | 0.910      | 45.786          | 0.000   | 45.831         | 0.000   | 24.996                |

Figure 7: Test for autocorrelation on daily log-returns from “adjusted closing” of CAC40 from Yahoo.com. Sample: 31-12-1994 to 31-12-2024.

By looking at the table above, the autocorrelation of the time series was tested using the Box–Pierce and Ljung–Box tests. For each lag starting  $k=3$ , we can reject the null hypothesis  $H_0$ : that the autocorrelation of a time is equal to 0 since the p-value for each test is smaller than the significance level  $\alpha=0.05$ . Therefore, the time series of the daily log-returns of the CAC40 is autocorrelated.

### Stylized Fact 7: Returns feature volatility clustering

Volatility clustering means that large price changes, i.e. returns with large absolute values or large squares, occur in clusters. For daily (and higher frequencies, say intra-day) large price changes tend to be followed by large price changes.

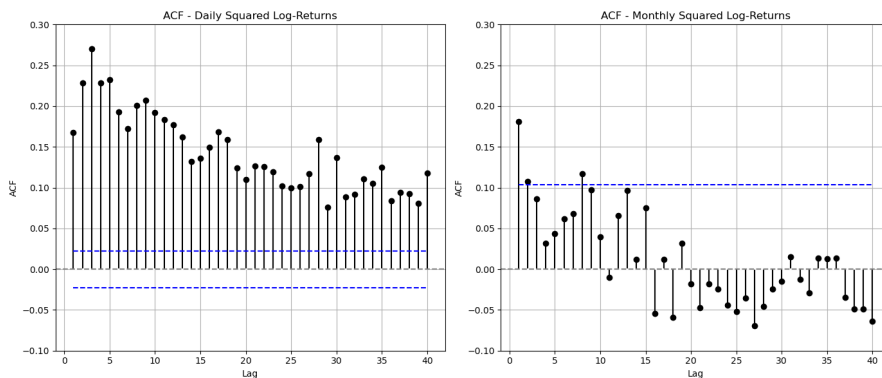


Figure 8: Time series of  $r_t$  the daily log-returns and squared log returns from the “adjusted closing” of CAC40 from Yahoo.com. Sample: 31-12-1994 to 31-12-2024.

The volatility clustering is reflected by significant autocorrelation of the squared returns:

$$\hat{C\hat{o}rr}(r_t^2, r_{t-k}^2) > 0 \quad \hat{C\hat{o}rr}(|r_t|, |r_{t-k}|) > 0 \quad \text{for } k \gg 0$$



Both squared ( $r_t^2$ ) and absolute ( $|r_t|$ ) daily returns exhibit significant autocorrelations. These autocorrelations are very persistent, that is they decay to 0 very slowly as  $k$  increases, indicating possible long-memory properties in higher moments of returns. Those autocorrelations become weaker and less persistent when the sampling interval is increased from a day to a week to a month.

As showed in the left graph ACFs remain positive and decay slowly, supporting the idea that high daily volatility today is likely to be followed by higher volatility in coming days. In the right panel (Monthly) the volatility clustering is weaker but still visible at monthly lags, reflecting that periods of market turbulence or calm can last multiple months.

### Stylized Fact 8: Leverage effect

Asset returns are negatively correlated with the changes of their volatilities: this negative correlation is called leverage effect. In particular there is strong empirical evidence that asset-return volatility rises after price declines, with larger declines inducing greater volatility spikes, and that:

$$\text{corr}(r_{t-j}, r_t^2) < 0; \text{ for } j > 0 \quad (1)$$

The CCF plot for the CAC 40 index clearly shows that the correlation coefficients at positive lags ( $j=1,2,3,\dots$ ) are significantly negative. This indicates that negative past returns are systematically followed by higher current volatility, whereas positive past returns are associated with lower volatility levels. Therefore, the empirical results confirm the presence of the Leverage Effect in the CAC 40 index.

In other words, Stylized Fact 8 is verified, as the relationship (1) holds true, demonstrating that negative shocks have a stronger and more persistent impact on market volatility than positive shocks of the same magnitude.

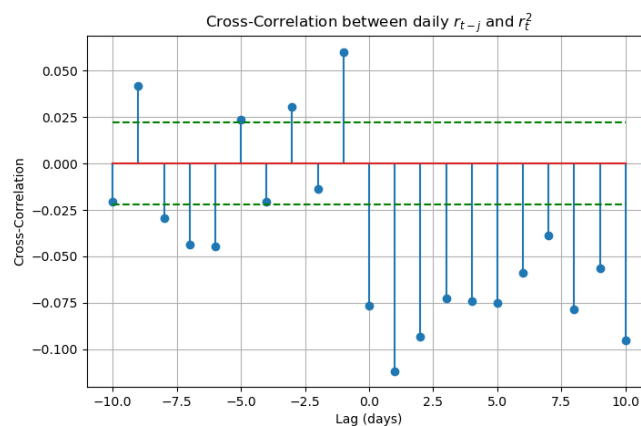


Figure 9: Empirical cross-correlation of daily lagged CAC40 log-returns, with squared returns. Daily log-return from the “adjusted closing” of CAC40 from Yahoo.com. Sample: 31-12-1994 to 31-12-2024. Dashed blue lines are the (asymptotic) bounds for the rejection region a significance test of each cross-correlation. A line above or below the blue dashed line represent a significant cross-correlation.

## Question 2

This section presents the comparative analysis of the Market Model (CAPM) and the Fama-French Three-Factor Model (FF3) on the monthly excess returns of the CAC 40 index  $R_{CAC40} - R_f$  over the full sample period.

### 2.I Summary Output of Regression Coefficients

| Coefficient                      | CAPM (Model 1) | FF 3-Factor (Model 2) |
|----------------------------------|----------------|-----------------------|
| Alpha ( $\alpha$ )               | -0.4383        | -0.4491               |
| Standard Error ( $\alpha$ )      | 0.2029         | 0.2035                |
| Beta Market ( $\beta_{Mkt}$ )    | 0.9189         | 0.9337                |
| Standard Error ( $\beta_{Mkt}$ ) | 0.0455         | 0.0470                |
| Beta SMB ( $\beta_{SMB}$ )       | N/A            | -0.0607               |
| Standard Error ( $\beta_{SMB}$ ) | N/A            | 0.0638                |
| Beta HML ( $\beta_{HML}$ )       | N/A            | 0.0427                |
| Standard Error ( $\beta_{HML}$ ) | N/A            | 0.0677                |
| Adjusted $R^2$                   | 0.5819         | 0.5815                |

Figure 10: Summary of Estimated Coefficients: CAPM vs. Fama-French 3-Factors (CAC 40 Monthly Excess Returns)

To begin our presentation of the results, we summarize the key results from the OLS regressions in the table below, presenting the estimated  $\alpha$  and  $\beta$  coefficients, along with their respective standard errors, for both the CAPM and the FF3 Model for direct

comparison.

### 2.II Fitted Regression Line of the Market Model

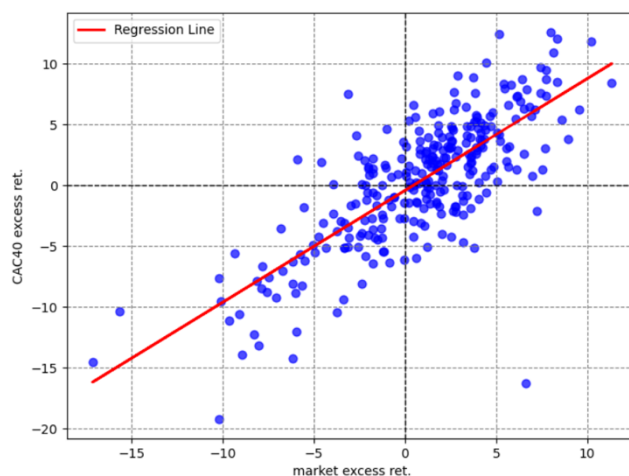


Figure 15: Scatter plot of the monthly excess return of the CAC40 from Yahoo.com. Sample: 31-12-1994 to 31-12-2024

The scatter plot below illustrates the monthly excess returns of the CAC 40 against the market excess return  $RMkt - R_f$ . The red line represents the estimated regression line from the CAPM (Model 1), which visually depicts the linear relationship defined by the estimated  $\alpha$  and  $\beta$ .

Visually, the data points show a strong linear tendency, clustering closely around the regression line. This confirms a and the market.

### 2.III Significance and Magnitude of Estimated Regression Coefficients

**Market Beta  $\beta_{Mkt}$**  : The Market Beta holds its position as the most statistically significant determinant of returns in both the CAPM and the FF3 model

**Magnitude:**  $\beta_{Mkt}$  is estimated near 0.92 (CAPM) and 0.93 (FF3). This value, slightly below 1.0, suggests the CAC 40 index is marginally less sensitive than the CRSP Value-Weighted market portfolio.

To determine if this slight difference from unity is statistically meaningful, we perform an hypothesis test for the Market Beta:

**Significance:** The absolute value of t-statistic for  $\beta_{\text{Mkt}}$  is  $\left| \frac{(0.9189-1)}{0.0455} \right| = 1,78$  (CAPM) and  $\left| \frac{(0.9337-1)}{0.0470} \right| = 1,41$  (FF3). This confirms that for both models, the CAC 40 index has an overall sensitivity to market movements that is equivalent to the broad market benchmark.

## 2.IV Comparison of the Goodness of Fit

The goodness of fit is measured by the Adjusted  $R^2$ :

- Model 1 (CAPM): Adj  $R^2 = 0.5819$
- Model 2 (FF 3-Factors): Adj  $R^2 = 0.5815$

The  $R^2$  for the FF3 model is slightly lower than the CAPM. This is unusual but confirms that the additional factors SMB and HML add no significant explanatory power for the CAC 40 index. For this asset, the CAPM is the more parsimonious model.

## 2.V Compatibility with Economic Theories

The compatibility of the models with economic theory hinges on the significance of the Alpha  $\alpha$ . We use the critical value of 1.96 for the 5% significance level.

### Compatibility with the Capital Asset Pricing Model (CAPM)

The CAPM is a single-factor equilibrium model that predicts  $\alpha$  should be statistically equal to zero.

- Result (Model 1 CAPM):  $\alpha = -0.4383$  (SE: 0.2029).
- t-statistic:  $\frac{-0.4383}{0.2029} = -2.16$ .

Since  $|t| > 1.96$ , the alpha is statistically significant at the 5% level. This significant negative Alpha rejects the strict CAPM hypothesis.

### Compatibility with the Arbitrage Pricing Theory (APT)

The APT posits that returns are explained by systematic factors, and that  $\alpha$  should be zero when all relevant factors are included.

- Result (Model 2 FF3):  $\alpha = -0.4491$  (SE: 0.2035).
- t-statistic:  $\frac{-0.4491}{0.2035} = -2.21$ .

Since  $|t| > 1.96$ , the Alpha is statistically significant at the 5% level.

Thus, the significant Alpha in both Model 1 and Model 2 indicates that neither the CAPM nor the standard Fama-French 3-Factor model is a sufficient specification for the CAC 40's returns over this period. This result contradicts the core prediction of both the CAPM and the APT (in its FF3 implementation) that the Alpha should be zero.